AI Meets Geometry
Lecture 3: Neural PDEs

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Overview

1. Physics-Informed Neural Networks (PINNs)
   - Losses
   - Architectures
2. Minimal Willmore Surfaces
   - The Willmore Conjecture
   - Neural Willmore Flow
3. Hands-on Python Coding
4. Summary

PINN Models

Launch Google Colab

EGD Website > Books > Download > Lecture 3 "Open in Colab"
`github.com/edhirst/XXIIEGD_AI_Meets_Geometry`

Neural Network Reminder

→ A neural network is a highly-parameterised nonlinear map f_θ built from linear affine layers and non-linear activation functions:

$$f_\theta := W_L \cdot \alpha(\cdots \alpha(W_1 x + b_1) \cdots) + b_L.$$

→ Training updates parameters to minimise a loss $\mathcal{L}$ function:

$$\theta \leftarrow \theta - \eta \, \nabla_\theta \mathcal{L}.$$

→ Universal approximation theorems motivate their use.

Physics-Informed Neural Networks

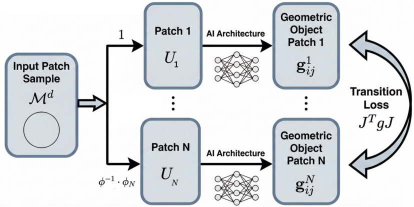
For PINNs, the loss $\mathcal{L}$ encodes the physics/geometry directly (no labelled output data required). PINN loss types:

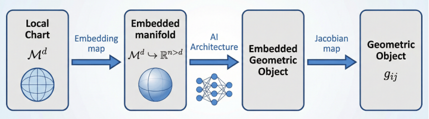
→ **Differential losses**: computed with automatic differentiation.
→ **Boundary losses**: boundary or initial conditions become extra penalty terms.
→ **Repeller losses**: deter convergence to known trivial solutions.

Geometric PINN Architectures

Manifold PINN Architectures

1) Atlas Architecture $\rightarrow$ Object is learnt independently in each patch,
...consistency between patches enforced with additional transition loss.
2) Embedding Architecture $\rightarrow$ Object is learnt globally on the embedding,
...but embedding must exist and be differentiable.





Immersion PINN Architectures

$\rightarrow$ Architecture is the immersion map, $f_\theta : \Sigma \longmapsto \mathbb{R}^3$.
$\rightarrow$ Loss is a scalar functional on this map, to be minimised.
$\rightarrow$ Training process reflects a Neural PDE equivalent of a geometric flow.

Willmore Conjecture

Definitions

For a closed surface Σ, with embedding $\varphi : \Sigma_{(u,v)} \hookrightarrow \mathbb{R}^3$.

$\rightarrow$ The first fundamental forms are:

$$E = \langle \partial_u \varphi, \partial_u \varphi \rangle, \qquad F = \langle \partial_u \varphi, \partial_v \varphi \rangle, \qquad G = \langle \partial_v \varphi, \partial_v \varphi \rangle.$$

$\rightarrow$ Defining the normal, $\hat{n} = \frac{\varphi_u \times \varphi_v}{|\varphi_u \times \varphi_v|}$, and area element $dA = \sqrt{EG - F^2} dudv$.

$\rightarrow$ The second fundamental forms are:

$$L = \langle \partial^2_{uu} \varphi, \hat{n} \rangle, \qquad M = \langle \partial^2_{uv} \varphi, \hat{n} \rangle, \qquad N = \langle \partial^2_{vv} \varphi, \hat{n} \rangle.$$

$\rightarrow$ The mean curvature is: $\quad H = \frac{EN - 2FM + GL}{2(EG - F^2)}$,

$\rightarrow$ Allowing definition of the Willmore Energy:

$$\mathcal{W}(\Sigma) := \iint_\Sigma H^2 dA \,.$$

The Problem

For Σ genus 1: $\mathcal{W}(\Sigma) \geq 2\pi^2$, *...with equality for the Clifford torus.*

...as proved via min-max theory by Marques–Neves 2012.

$\rightarrow$ For genus 0 $\mathcal{W}(\Sigma)$ is minimised by the round sphere.

$\rightarrow$ For genus $\geq 2 \Rightarrow$ open problem. (conjectured Lawson surfaces)

Problem Setup

Use the Immersion PINN Architecture to learn the embedding map of the minimal Willmore torus.

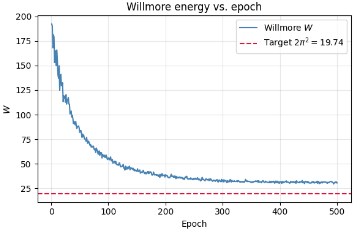
→ Sample the torus fundamental domain: $[0, 2\pi] \times [0, 2\pi]$.
→ Design a simple initial embedding in $\mathbb{R}^3$.
→ Supervised learn this embedding to set topology.
→ Build the PINN Willmore and regularity losses.
→ Run the PINN training to find the minimal surface.

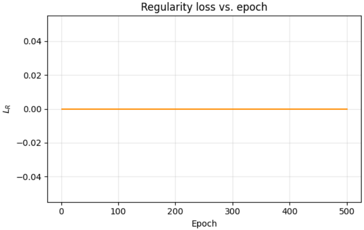
Running the Code

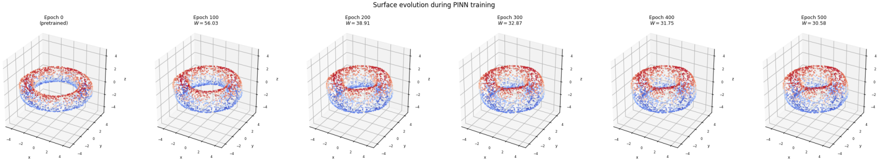
Return to Colab: `github.com/edhirst/XXIIEGD_AI_Meets_Geometry`
In the jupyter notebook `lecture_3.ipynb`.

→ Read each cell, then run it.
→ Follow cells sequentially until the end.
→ Return to the Exercise cells, write your own `python` code to complete them.

Summary







Summary

$\rightarrow$ PINN architectures model geometric objects with AI models.
$\rightarrow$ Training objectives become geometric functional minimisation.
$\rightarrow$ Solutions guide insight, uncover symmetry, and can lead to rigorous proofs.